DupeCheck: Business Requirements Document (BRD)

Field	Detail
Product	DupeCheck, a duplicate test alert for clinicians at the moment of ordering
Document	Business Requirements Document, v0.1 DRAFT
Author	Rohan Singh, Product Owner
Domain	US healthcare, payer and health services, clinical decision support
Status	Draft for review. Items marked **[Validate]** need confirmation from stakeholders or real data
Data policy	Synthetic data only. No real patient data (PHI) in any artifact

1. Executive summary

When patients move between hospitals, clinics and labs, a clinician ordering a test often cannot see that the same or a very similar test was recently done elsewhere. The test is repeated. That costs money, adds patient burden (time, contrast, radiation for imaging) and delays care.

DupeCheck is a calm, explainable alert shown at the moment of ordering: "A similar test was done 21 days ago at another facility. Here is the result." The clinician always makes the final decision.

The proposal in one line: use transparent rules in v1, draw on both clinical data and payer claims data to see tests done at other facilities, and measure avoided repeats and avoided cost.

Why this belongs in a payer and data analytics context: a payer sees services across all of a member's providers, which is exactly the visibility an individual hospital lacks. This is a hypothesis to validate, not a settled fact (see 4.3 and 11).

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2. Problem framing

2.1 Problem statement

Ordering clinicians frequently lack visibility of tests already performed at other facilities, so avoidable duplicate tests are ordered. This increases cost and patient burden, and no routine, low-friction safeguard exists at the point of ordering.

2.2 Who is affected

Group	Effect
Patients	Repeat procedures, extra radiation or contrast exposure, wasted time, out of pocket cost
Ordering clinicians	Decide without full history, spend time chasing prior reports
Health plans and payers	Pay for avoidable tests, higher medical cost
Hospitals and labs	Wasted capacity, delayed throughput, billing disputes

2.3 Current state versus desired state

Dimension	Today	Desired

2.4 Root causes (5 whys summary)

1. Duplicate test ordered, because the clinician did not know a prior test existed.
2. Did not know, because the prior result sits in another organization's system.
3. Not shared, because systems are fragmented and exchange is inconsistent.
4. Exchange is inconsistent, because incentives, standards adoption and patient matching vary.
5. No safeguard at order entry, because existing tools show records on request rather than pushing a warning at the moment of decision.

2.5 Evidence base

Confidence reflects source quality. Several figures come from industry surveys or secondary summaries, so **verify against the original source before quoting externally**.

#	Finding	Source type	Confidence	Caveat
E1	Nearly one fifth of patient records in an organization were duplicates, on average	Black Book survey, about 1,400 health technology managers, 2017 to 2018	Low to medium	Industry survey, self reported
E2	Duplicate records cost the US system over $6 billion a year, about $1,950 per inpatient stay and over $800 per ED visit	Same survey, reported by Healthcare Finance News	Low to medium	Survey estimate, dated
E3	Hospitals using master patient index tools reported far better matching than those without (93 percent of registrations and 85 percent of externally shared records, versus 24 percent match rate without tools)	Same survey	Low to medium	Supports identity matching as a key risk
E4	Duplicate radiology tests fell significantly when hospitals shared data through health information exchange	Ayabakan et al., MIS Quarterly 2017, summarized by Temple University	Medium	Study focused on heart failure patients
E5	Estimated duplicate radiology cost of about $300 per heart failure patient, roughly $1.12 million in that study; a modelled estimate of up to $77.8 billion annual net HIE savings was cited	Same summary	Low for the large figure	Modelled upper bound, treat with caution
E6	VA providers who used a shared record viewer were about five percentage points less likely to order duplicate imaging	Retrospective VA and DoD study (FY2018), summarized by The Incidental Economist	Medium to high	Could not separate necessary from unnecessary repeats
E7	Incomplete record transfer between incompatible electronic systems led to potentially costly duplicate testing in transferred patients	PMC preliminary study, 85 patients, 2006 to 2007	Low	Small and dated
E8	Only 43 percent of US acute care hospitals routinely participated in key interoperability functions in 2023	Advocacy report citing federal research	Low	Verify against ONC data

Reading the evidence honestly: the problem is real and repeatedly documented, and information sharing reduces duplicates. What is thin is current, rigorous data on how many repeats are truly avoidable. That is why the first phase includes a baseline measurement, and why alerts never block care.

2.6 Opportunity sizing model

Avoided annual spend = Eligible orders x Duplicate rate x Avoidable share x Alert action rate x Average test cost

Illustrative worked example only. Every input is a placeholder to be replaced with Carelon data [Validate].

Input	Placeholder
Eligible imaging and lab orders per year in pilot population	500,000
Duplicate rate (similar test within window)	3 percent
Avoidable share (no clinical reason to repeat)	50 percent
Alert action rate (clinician cancels or modifies)	40 percent
Average test cost	$250
Avoided spend per year	**about $750,000**

Sensitivity: the result is most sensitive to duplicate rate and avoidable share, so Phase 0 measures both from real, de-identified data.

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3. Business objectives and success measures

ID	Objective	Measure	Baseline	Target (illustrative)
BO-1	Quantify avoidable duplicate testing	Duplicate rate by test type	Not measured **[Validate]**	Baseline established in Phase 0
BO-2	Reduce avoidable duplicate orders	Avoidable duplicates per 1,000 eligible orders	Phase 0	20 to 30 percent relative reduction at pilot sites
BO-3	Keep alerts useful, not noisy	Alert action rate; override rate	Not applicable	Action rate at or above 30 percent; override reasons reviewed monthly
BO-4	Reduce unnecessary spend	Estimated avoided cost	Phase 0	Validated with finance and clinical leads
BO-5	Maintain safety and trust	Zero blocked urgent orders; clinician satisfaction	Not applicable	100 percent urgent orders unblocked; satisfaction survey at pilot end

4. Requirement gathering

4.1 Approach

Method	Purpose	Participants	Output
Stakeholder interviews (45 min)	Understand workflow, pain, constraints	Ordering clinicians, radiology and lab leads, quality and patient safety, claims and utilization management, data engineering, compliance	Interview notes, validated needs

Method	Purpose	Participants	Output
Workflow observation or walkthrough	See how ordering really happens	2 or 3 clinicians	Current state flow
Data profiling	Size the problem on real de-identified data	Data and analytics team	Duplicate rate, test mix, claims latency
Competitive and standards scan	Understand EHR and exchange features already in market	Product and architecture	Differentiation view
Requirements workshop	Prioritize and agree scope	All key stakeholders	Signed off scope and MoSCoW list

4.2 Interview guide (core questions)

Ordering clinicians

1. Walk me through the last time you ordered an imaging study. What did you check first?
2. How do you find out whether a patient already had a similar test elsewhere?
3. How often do you discover a repeat after the fact? What happened?
4. What would make an alert helpful versus annoying? What do you ignore today?
5. When is repeating a test justified even if one was recent?

Radiology and lab leads

6. How do you handle outside images and results today? How long does it take?
7. Which tests are most commonly repeated unnecessarily?

Quality, patient safety and compliance

8. What rules govern showing external patient data at order time?
9. What audit and consent requirements apply?

Claims, utilization and data teams

10. How quickly do claims for imaging and labs become visible, and how complete are they across providers?
11. Can we get de-identified data to measure duplicate rates?

4.3 Hypotheses to validate

ID	Hypothesis	How to test	Status
H1	Avoidable duplicates are frequent enough to matter at pilot scale	Data profiling	Open
H2	Clinicians will act on a calm alert at order time	Prototype sessions, pilot metrics	Open
H3	Payer claims can reveal tests done at other facilities, even without clinical records	Data profiling, check claims lag	Open
H4	Claims latency (days to weeks) is short enough to catch most repeats inside the review windows	Data profiling	Open
H5	The alert can be delivered inside the EHR via standard integration	Architecture review	Open

4.4 Assumption log

ID	Assumption	Owner to confirm
A1	Pilot clinicians use an EHR that supports CDS Hooks or an equivalent decision support integration	Architecture
A2	De-identified claims and result data is available for baseline analysis	Data governance
A3	Review windows per test type are clinically defensible once validated	Clinical advisory
A4	Patient matching quality is sufficient to avoid wrong alerts	Data engineering

5. Scope

In scope (v1): duplicate detection at diagnostic test ordering for selected imaging and lab tests; rules based matching using test code, or modality plus body region, within configurable windows; inclusion of external results; one calm alert with prior test details; clinician actions with override reasons; audit trail; admin insights.

Out of scope (v1): automated ordering decisions; patient facing features; claims adjudication or payment denial logic; predictive models; free text report reading; replacing any EHR.

Phase gate for AI: AI features (report reading, test mapping, smarter alerting) move forward only after v1 proves value and passes clinical safety review.

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6. Business requirements

ID	Requirement	Priority	Linked objective
BR-01	The business needs a baseline measure of duplicate tests by test type from real de-identified data	Must	BO-1
BR-02	Ordering clinicians must be warned at order time when a similar test was done recently, including at other facilities	Must	BO-2
BR-03	The warning must explain why it fired and show the prior test, date, source and result summary	Must	BO-3
BR-04	The clinician must always be able to proceed, and urgent orders must never be blocked	Must	BO-5
BR-05	Overrides must capture a reason and be auditable	Must	BO-3, BO-5
BR-06	The business must be able to tune windows and rules without a code release, under clinical governance	Should	BO-3
BR-07	Leadership must see alerts, outcomes, avoided tests and estimated avoided cost	Must	BO-4
BR-08	Access must follow minimum necessary and be role based, with full audit logging	Must	BO-5
BR-09	The solution should combine clinical exchange data and payer claims data to widen visibility	Should	BO-2
BR-10	The design must allow later AI features without rework of the core service	Could	BO-2

7. Business use cases

ID	Actor	Scenario	Outcome
UC-1	Ordering clinician	Orders CT chest; similar CT was done 21 days ago at another hospital	Alert shown, clinician reviews prior result, cancels order
UC-2	Ordering clinician	Orders HbA1c; last one was 140 days ago, outside the window	Informational note only, order proceeds
UC-3	Ordering clinician	Orders urgent repeat imaging for a worsening patient	Alert shown, order proceeds with a reason, care not delayed
UC-4	Quality lead	Reviews override reasons and alert acceptance monthly	Rules refined, noise reduced
UC-5	Clinical governance	Approves or edits review windows by test type	Rules updated with audit trail
UC-6	Analytics lead	Tracks avoided tests and estimated spend	Business case validated
UC-7	Compliance officer	Audits who viewed external results and why	Audit readiness

8. Stakeholders

Stakeholder	Interest	Role
Product management leadership	Value, roadmap	Sponsor and approver
Ordering clinicians	Usable, accurate alerts	Primary users
Radiology and lab leaders	Throughput, outside results	Consulted
Quality and patient safety	Safety, trust	Consulted, approver for rules
Claims and utilization management	Spend, data availability	Consulted
Data and analytics engineering	Pipelines, matching	Responsible for delivery
Compliance and privacy	HIPAA, audit	Approver
Product Owner (Rohan Singh)	Requirements, backlog, outcomes	Accountable

9. Constraints and dependencies

- HIPAA minimum necessary access, encryption, audit logging, and a signed BAA for any cloud service.
- Integration into the clinician's existing EHR workflow, rather than a separate tool.
- Availability and quality of external results and claims feeds.
- Patient identity matching across systems.
- Clinical sign off on review windows and match logic.

10. Risks

Risk	Impact	Likelihood	Mitigation
Alert fatigue, clinicians ignore alerts	High	Medium	One calm alert per order, tune windows, monitor action rate
Wrong patient match produces a wrong alert	High	Medium	Conservative match thresholds, show identity evidence, clinician verifies

Risk	Impact	Likelihood	Mitigation
Claims data arrives too late to catch repeats	Medium	Medium	Combine with clinical exchange data, measure latency in Phase 0
Alert seen as blocking care	High	Low	Never hard stop, urgent orders unblocked
Software treated as a medical device	Medium	Low to medium	Keep to transparent rule based support with clinician in the loop, regulatory review
Duplicate rate lower than assumed	High	Medium	Phase 0 baseline before commitment, decision gate
Bias against clinicians (seen as policing)	Medium	Medium	Position as support, report at group level not individual blame

11. Differentiation (honest view)

Health information exchanges and EHR vendor features already let clinicians view outside records on request. DupeCheck does not try to replace them. Its value is a **push warning at the moment of ordering**, with explainable rules and measurable outcomes, and a possible **claims based view across facilities** that a single provider cannot see. This differentiation is a hypothesis (H3, H4) to be tested.

12. High level timeline

Phase	Weeks	Output
0 Discovery and baseline	1 to 6	Interviews, data profiling, validated scope, decision gate
1 MVP pilot	7 to 18	Rules engine, alert in one pilot workflow, audit, insights
2 Expansion	19 to 30	More tests and sites, claims integration, rule tuning
3 Intelligence (gated)	31 onward	NLP report reading, test mapping, smarter alerting

13. Open questions

1. Which line of business, facilities or EHR will the pilot use?
2. What data (claims, exchange, results) is available now and who owns it?
3. Who in clinical governance approves review windows?
4. Are there existing duplicate or imaging utilization programs this should extend?
5. What counts as success for leadership in the first two quarters?

14. Approvals

Name	Role	Decision	Date
To be assigned	Sponsor	Pending	
To be assigned	Clinical lead	Pending	
To be assigned	Compliance	Pending	

15. Sources

- Healthcare Finance News, "Patient matching: Hospitals can save millions by improving it" (Black Book survey): https://www.healthcarefinancenews.com/news/patient-matching-hospitals-can-save-millions-improving-it
- Healthcare IT News, "Duplication, fragmentation hamper interoperability efforts, impact patient safety": https://healthcareitnews.com/node/545870

- The Incidental Economist, "Integrated Health Record Viewers and Duplicate Imaging": https://theincidentaleconomist.com/wordpress/recent-research-integrated-health-record-viewers-and-duplicate-imaging/
- PMC, "A preliminary look at duplicate testing associated with lack of electronic health record interoperability for transferred patients": https://pmc.ncbi.nlm.nih.gov/articles/PMC2995707
- Temple University MIS community summary of Ayabakan et al., MIS Quarterly 2017 (original paper should be consulted before quoting)